

Actividad 5 - Docker Compose

1. Despliegue de Let's Chat mediante Docker Compose

Creamos el fichero docker-compose.yml con nano y lo levantamos en modo daemon. Docker Compose descarga las imágenes necesarias (lets-chat y mongo) y arranca los contenedores:

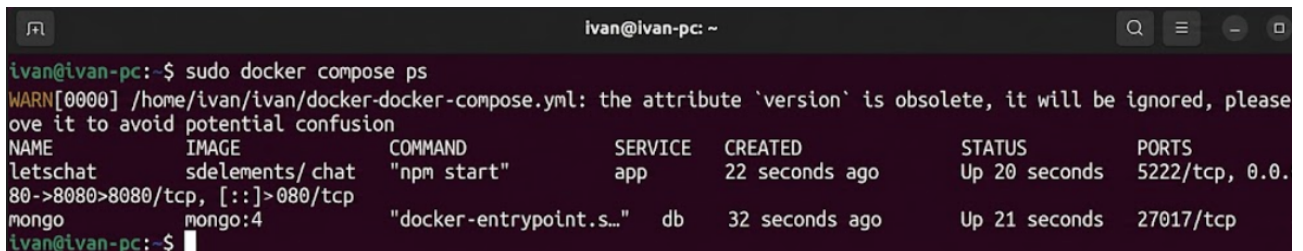
```
sudo nano docker-compose.yml
sudo docker compose up -d
```



```
ivan-cliente@ivan-cliente: ~
ivan-cliente@ivan-cliente:~$ sudo nano docker-compose.yml
ivan-cliente@ivan-cliente:~$ sudo docker compose up -d
WARN[0000] /home/ivan-cliente/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 26/26
✓ Image sdelements/lets-chat Pulled
✓ Image mongo:4 Pulled
✓ Network ivan-cliente_default Created
✓ Volume ivan-cliente_mongo Created
✓ Container mongo Started
✓ Container letschat Started
ivan-cliente@ivan-cliente:~$
```

Verificamos que los contenedores están corriendo:

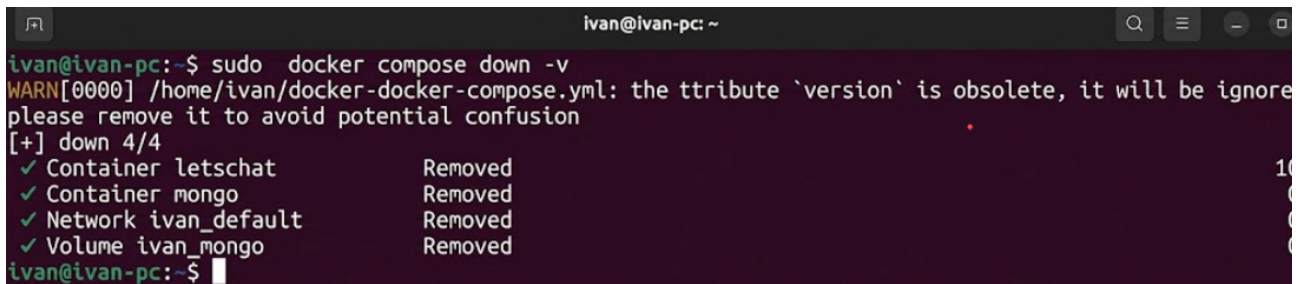
```
sudo docker compose ps
```



```
ivan@ivan-pc: ~
ivan@ivan-pc:~$ sudo docker compose ps
WARN[0000] /home/ivan/ivan/docker-docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
NAME          IMAGE          COMMAND          SERVICE  CREATED          STATUS          PORTS
letschat      sdelements/ chat "npm start"      app      22 seconds ago   Up 20 seconds   5222/tcp, 0.0.0.0->8080->8080/tcp, [::]:8080->8080/tcp
mongo         mongo:4        "docker-entrypoint.s..." db        32 seconds ago   Up 21 seconds   27017/tcp
ivan@ivan-pc:~$
```

Eliminamos la aplicación multicontenedor junto con los volúmenes usando el flag -v:

```
sudo docker compose down -v
```



```
ivan@ivan-pc: ~
ivan@ivan-pc:~$ sudo docker compose down -v
WARN[0000] /home/ivan/docker-docker-compose.yml: the ttribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] down 4/4
✓ Container letschat Removed
✓ Container mongo Removed
✓ Network ivan_default Removed
✓ Volume ivan_mongo Removed
ivan@ivan-pc:~$
```

2. Despliegue de Temperaturas mediante Docker Compose

Editamos el docker-compose.yml con la receta para la aplicación Temperaturas. El fichero define dos servicios (backend y frontend) conectados a red_temperaturas, con depends_on para que el frontend espere al backend:

```
sudo nano docker-compose.yml
```

```
GNU nano 7.2                                docker-compose.yml *
version: "3.8"

services:
  backend:
    container_name: temperaturas-backend
    image: iesgn/temperaturas_backend
    restart: always
    networks:
      - red_temperaturas

  frontend:
    container_name: temperatur-s-frontent
    image: iesgn/temperaturas_frontend
    restart: always
    depends_on:
      - backend
    networks:
      - red_temperaturas
    ports:
      - 80:3000

networks:
  red_temperaturas:
ivan@ivan:~$
```

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^X Salir ^R Leer fich. ^\ Reemplazar ^U Pegar ^J Justificar ^/ Ir a línea M-E Rehacer M-6 Copi

Lo levantamos y comprobamos que ambos contenedores están en marcha, luego los eliminamos:

```
sudo docker compose up -d
sudo docker compose ps
sudo docker compose down -v
```

```
ivan@ivan-pc:~$ sudo docker compose ps
WARN[0000] /home/ivan/docker-docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please
ove it to avoid potential confusion
NAME                IMAGE                                COMMAND                                SERVICE  CREATED          STATUS
S
temperaturas-backend iesgn/temperaturas_backend          "python3 app.py"                     backend  23 seconds ago  Up 22 seconds
/tcp
temperaturas-frontend iesgn/temperaturas_frontend        "python3 app.py"                     frontend 22 seconds ago  Up 21 seconds
0.0:80->3000/tcp, [::]:80->3000/tcp
ivan@ivan-pc:~$ sudo docker compose down -v
WARN[0000] /home/ivan/dockocker/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please
ove it to avoid potential confusion
[+] down 3/3
✓ Container temperaturas-frontend      Removed
✓ Container temperaturas-backend       Removed
✓ Network ivan_red_temperaturas        Removed
ivan@ivan-pc:~$
```

3. Despliegue de WordPress + MariaDB mediante Docker Compose

Creamos un fichero aparte llamado docker-compose.wp+mariadb.yml. Define los servicios servidor_mysql y servidor_wp con sus volúmenes, variables de entorno y la red red_wp de tipo bridge:

```
sudo nano docker-compose.wp+mariadb.yml
```

```
ivan@ivan-pc: ~
GNU nano 7.2 docker-compose.wp+mariadb.yml *
version: '3.8'

services:
  servidor_mysql:
    image: mariadb
    container_name: servidor_mysql
    networks:
      - red_wp
    volumes:
      - /opt/mysql_wp:/var/lib/mysql
    environment:
      MYSQL_DATABASE: bd_wp
      MYSQL_USER: user_wp
      MYSQL_PASSWORD: prueba
      MYSQL_ROOT_PASSWORD: prueba

  servidor_wp:
    image: wordpress
    container_name: servidor_wp
    networks:
      - red_wp
    volumes:
      - /opt/wordpress:/var/www/html/wp-content
    environment:
      WORDPRESS_DB_HOST: servidor_mysql
      WORDPRESS_DB_USER: user_wp
      WORDPRESS_DB_PASSWORD: prueba
      WORDPRESS_DB_NAME: bd_wp

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^X Salir      ^R Leer fich. ^\ Reemplazar ^U Pegar      ^J Justificar ^/ Ir a línea  M-E Rehacer  M-6 Copia
```

Lo levantamos especificando el fichero con el flag -f y verificamos que los contenedores están corriendo:

```
sudo docker compose -f docker-compose.wp+mariadb.yml up -d
sudo docker ps
```

```
ivan@ivan-pc: ~
ivan@ivan-pc:~$ sudo docker compose -f docker-compose.wp+mariadb.yml up -d
WARN[0000] /home/ivan/docker-docker-compose.wp+mariadb.yml: the attribute `version` is obsolete, it will be ignored
please remove it to avoid potential confusion
[+] up 3/3
✓ Network ivan_red_wp Created
✓ Container servidor_mysql Started
✓ Container servidor_wp Started
ivan@ivan-pc:~$ sudo docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
NAMES
32981bb1750e   wordpress     "docker-entrypoint.s..." 22 seconds ago Up 19 seconds  0.0.0.0:80->80/tcp, [::]:80->80/tcp
servidor_wp
28689c1bcb8f   mariadb       "docker-entrypoint.s..." 22 seconds ago Up 21 seconds  3306/tcp
servidor_mysql
ivan@ivan-pc:~$
```

Accedemos a 127.0.0.1 en el navegador y comprobamos que el instalador de WordPress está disponible:

